## 🔗 Topic 2 at a glance: how the three papers connect

- **Function adapts to form** more than form follows function: the tongue compensates for malocclusion in speech (Johnson and Sandy) and in swallowing (Proffit and Mason).
- **Duration is what moves teeth:** sucking habits lasting years change the arches (Warren and Bishara), and the resting tongue matters, while a brief swallow or a spoken sound does not.
- **Correlation is not causation:** speech defects, tongue thrust and bottle-feeding all co-exist with malocclusion without proving they cause it.
- **Clinically:** treat teeth for teeth and speech with a speech therapist; stop pacifier and digit habits by about 2–3 years; do not treat tongue thrust before puberty.

# Tooth Position and Speech: Is There a Relationship?

by **Johnson and Sandy** · Angle Orthod 1999;69(4):306–310 · Review article, University of Bristol

## 🎭 Study purpose

To review whether **tooth position (malocclusion) affects speech**, and whether **changing tooth position improves speech**, and to explain why this is so hard to study. Narrative review of **42 references**; no search strategy or pooled analysis.

## 🗣️ Background (p 306)

- Speech uses 3 systems: **respiratory, phonatory, articulatory**. Teeth, lips and tongue articulate consonants by blocking or shaping airflow.
- Articulatory disorders are not the most severe, but make up **50–60 %** of all speech disorders.
- Many people speak normally despite abnormal tooth position, which points to **compensation**.
- If teeth did affect speech, a speech disorder could become an **indication** for orthodontics and change the **timing** of treatment.

## 🔗 Harvold's three possible links (p 306)

| # | Mechanism | What it would mean (derived) |
|---|---|---|
| 1 | **Coincidence:** an occlusal or skeletal problem and, by chance, an articulation problem | Correcting the teeth will not change speech |
| 2 | **Common cause:** a genetic or metabolic CNS disorder gives poor motor control *and* distorted morphology | Both appear together; neither causes the other |
| 3 | **True cause and effect:** occlusal or structural anomalies impair articulation | The only case in which orthodontics could help speech |

## ⚠️ Why the research is hard (p 309)

**3 study types:** 1. speech patients examined for malocclusion; 2. malocclusion patients examined for speech; 3. unselected populations studied for both. Only type 3 avoids selection, but many of those were done on undergraduate students (Oliver & Evans, Howell, Laine: dental students), which may introduce bias.

- **No animal model:** speech is unique to humans.
- **Definitions vary** for articulation defects, distortions and disorders, and for malocclusion; **reliability and validity are seldom reported**.
- **Language:** Laine's work was Finnish; he found the same sounds distorted as in other languages, but cross-population comparison stays difficult.
- **Phonetic context:** lip, incisor and tongue positions for /s/ differ within and between speakers depending on the surrounding sounds, which may explain conflicting results (p 307).
- **Lateral dimensions:** measuring points differ and palatal size shows sexual dimorphism (p 308).
- **Compensation** is enormous and hides any effect of tooth position.

## 📊 Key studies: malocclusion in general (p 306–308)

| Study | Sample | Finding |
|---|---|---|
| Fymbo | 410 students, relatively unselected | No meaningful analysis; trends only: malocclusion gave more trouble with dental sounds, and speech severity varied with dental severity |
| Frowine & Moser | Small case series | Satisfactory speech whatever the malocclusion |
| Rathbone | Small sample | Related "without question", but severity of malocclusion did not match severity of speech defect |
| Hopkin & McEwen | 101 children at speech-therapy clinics | Defects **as likely with normal occlusion** as with malocclusion |
| Jann et al. (randomized longitudinal) | Elementary school children, grade 1 → 5 | Articulation variations **62 %** → **71 %**; malocclusions **23 %** → **71 %** |
| Oliver & Evans | 35 dental students (arch width, length, palatal height, volume) | Poor articulators had **more "normal"** dental features |
| Howell (kinesiograph) | Nearly 100 dental students | Elongated movement envelope in every malocclusion; 4 patterns matched incisor class; **no articulation problems** in the group |
| Laine series | 451 dental students, Finland | Unselected; same sounds distorted as in other languages |
| Longitudinal study (ref 32) | School children 9–11 years | Neither dental stage nor age related to articulation disorders |

## 🔍 Evidence by occlusal trait (p 307–308)

| Trait | Key findings | Verdict |
|---|---|---|
| **Anterior open bite** | Fymbo: **63 %** defective speech, **4 %** superior; Bernstein (437 + matched controls): only open bite strongly linked, to **lisping**, but lisp severity ≠ open-bite size; Pomerantz & Zeller: open bite or edge-to-edge linked to /s/ /z/ /th/ /l/; Laine: consonants made too far forward **×3.4** | **Most often implicated**; alone usually mild, worse with Class III |
| Class III (mesial occlusion) | Low, flaccid tongue fails to constrict the linguoalveolar valve for sibilants; Guay (12 adolescents, full anterior crossbite, cephalometric): tongue low and retruded, normal /s/ in only 1; Laine: **×4.5** mesial occlusion, **×3.7** mandibular overjet, perhaps from oropharyngeal volume; Bloomer: mild, even considerable, discrepancy may cause no problems | Link; worse when combined with open bite |

| Trait | Key findings | Verdict |
|---|---|---|
| Increased overjet (Class II) | Tongue adapts in Class II div 1 (Subtelny, Jensen); Benediktsson: mandible protruded more for /s/; Blyth (200 orthodontic children): skeleton had little effect on interdental sigmatism; Laine: /s/ distortion, sounds too far forward; no link for postnormal buccal segments | Link to /s/ |
| Deep overbite | Lubit (300 consecutive): /s/ link but negligible overall; Ingervall & Sarnas: lateral lisp; Laine: none | Conflicting |
| Lateral crossbite, narrow palate | Laine: **×1.7**; reduced maxillary size and narrow palate implicated; mandibular arch size not linked; others dispute | Weak |
| Upper spacing, missing upper incisors | Laine: /l/ /n/ /d/ /r/; many compensate; none for lower spacing or crowding | Link, but compensation common |

## 🦷 Does orthodontic treatment change speech? (p 308)

Removable appliances change speech **transiently**, like new dentures (not reviewed further). For corrective treatment there is a surprising paucity of literature.

| Study | Sample | Finding |
|---|---|---|
| Kessler | Clinical observation | Speech often improves once the occlusion is corrected |
| Rathbone & Snidecor | 8 patients, 4 years of orthodontics, no speech therapy | Faulty sounds **6.4** → **1.5** (mean); fricatives /s/ /z/ /sh/ /zh/ stayed faulty |
| Prospective RCT | 25 patients with functional voice disturbances, treatment vs placebo | Occlusal adjustment removing all interferences improved voice at **2 months**; placebo unchanged |

The RCT tested **voice**, not articulation, and **occlusal adjustment**, not orthodontics (derived).

## 🔪 Orthognathic surgery and speech (p 309)

Complex, because surgery moves both teeth and jaws.

| Study | Sample | Finding |
|---|---|---|
| Higley | 2 case reports, mandibular surgery | One gained intelligibility; the other swapped an anterior for a lateral sigmatism |
| Goodstein | 5, mandibular length reduced | No change |
| Ref 38 | — | Errors may **increase** as the jaw takes its new position |
| Dalston & Vig | 40 women | **No postoperative change** |
| Ruscello | Review | Surgery alters the articulators and improves distorted speech |
| Yamaguchi | 15 adult Japanese, mandibular prognathism, videotaped | Most clearly improved; **no statistics** |

Evidence equivocal: no guarantees.

## ⚖️ Bloomer: adaptation decides (p 309)

| Structure | Function | Speech |
|---|---|---|
| Normal | Normal | Normal |
| Abnormal | Adaptive | Normal |
| Abnormal | No adaptation | Abnormal |
| Normal | Abnormal | Abnormal |

## 🧠 Main conclusions

1. Many traditional occlusion–speech links have no scientifically sound basis.
2. Most consistent traits: **Class III, anterior open bite, increased overjet, spacing**.
3. Severity of malocclusion does not predict severity of misarticulation (Fymbo saw a link; Rathbone and Bernstein did not).
4. Sounds learned last in childhood are the ones most often distorted, perhaps reflecting their difficulty.
5. Ability to adapt plays a significant role (Bloomer, above).
6. No substantial evidence that orthodontics influences articulation disorders: **co-operation with a speech therapist is essential**.

## 💡 Clinical implications for the orthodontist

- **Listen at the first visit:** misarticulated sounds can be recognised in normal conversation; note them and any link to the malocclusion (p 309).
- **Highest-risk pattern:** anterior open bite with Class III (mesial occlusion) gives more severe misarticulation (p 308).
- **Do not promise speech change** from orthodontics or orthognathic surgery; record speech before treatment so later change is not blamed on it (derived).
- **Warn surgical patients** that errors may increase after surgery as the jaw takes its new position (p 309).
- **Warn about removable appliances:** transient speech change, as with dentures (p 308).
- **Speech defect plus malocclusion:** co-operation with a speech therapist is essential; speech alone is not an indication or a reason to start early (derived from p 306, 309).

| ✅ Number | Meaning |
|---|---|
| 50–60 % | articulatory share of all speech disorders |
| 63 % vs 4 % | open bite: defective vs superior speech (Fymbo) |
| ×4.5 / ×3.7 / ×3.4 / ×1.7 | risk of consonants made too far forward: mesial occlusion / mandibular overjet / open bite / lateral crossbite (Laine) |
| 437 | Bernstein's speech-problem children: only open bite linked (to lisping) |
| 62 → 71 % vs 23 → 71 % | articulation variations vs malocclusions, grade 1 → 5 (Jann) |
| 6.4 → 1.5 | faulty sounds before and after orthodontics, 8 patients |
| 40 | Dalston & Vig: women with no speech change after surgery |
| 25 / 2 months | RCT: occlusal adjustment improved voice, placebo did not |

# Duration of Nutritive and Nonnutritive Sucking Behaviors and Their Effects on the Dental Arches in the Primary Dentition

by **Warren and Bishara** · Am J Orthod Dentofacial Orthop 2002;121(4):347–356 · University of Iowa

## 🎯 Study purpose

To find whether **how long** a child breast-feeds (nutritive sucking) or uses a **pacifier or digit** (nonnutritive sucking) changes the dental arches and occlusion at the **end of the primary dentition**.

## 📖 Background

- Harm from prolonged thumb or finger sucking was described as early as the **1870s** (Campbell; Chandler).
- **Breast-feeding** evidence was thin: Meyers and Hertzberg (454 Massachusetts children, 10–12 years) found a weak, nonsignificant link between bottle-feeding and malocclusion; the 1981 US National Health Interview Survey (3–18 years) linked longer breast-feeding to less malocclusion. Both relied on parents' reports of orthodontic treatment, not examination. Bishara's longitudinal study of **122 Iowa infants** found no arch differences among 6 feeding and sucking groups at 18 or 36 months.
- **Nonnutritive sucking** is better studied: Kohler and Holst (Swedish 4-year-olds) found more open bite, excess overjet and posterior crossbite; Adair (218 US children, 2–4 years) found pacifier users had more overjet, Class II canines and molars, open bite and crossbite; Ögaard (445 3-year-olds) found pacifiers narrowed the maxillary and widened the mandibular intercanine width; narrowing needed about **2 years** of use, widening **3 years**.
- Most were **cross-sectional**, so the role of habit **duration** was unclear.

## 🔬 Study design

- **Prospective birth cohort** (Iowa, mothers recruited in postpartum wards March 1992–January 1995, nearly 700 children). Mailed parent questionnaires at 3, 6, 9, 12, 16 and 20 months on breast- and bottle-feeding and regular sucking of pacifier, thumb, finger or objects; sucking-habit questions continued at 24 months, then yearly.
- Habit type (primarily pacifier or digit) taken from the **last report**; 45 children unclassifiable or reporting both.
- Duration groups by whether the habit was present at 12, 24, 36 and 48 months: **<12, 12–24, 24–36, 36–48, ≥48 months**. Children with <12 months of nonnutritive sucking (n = 119) were regrouped by breast-feeding: none, <6, 6–12, >12 months.
- Exam at **4.5–5 years**: alginate impressions and wax bites from 547 children; 98 excluded (a permanent tooth present or erupting), 56 excluded (missing habit data): **372 analysed** (as printed; 547 − 98 − 56 = 393).
- Warren scored canine Angle class, crossbites and open bite; Bishara marked landmarks. Calipers to **0.05 mm**; each measure taken at least twice by 2 examiners, a third time if they differed by >0.5 mm.
- 1-way ANOVA (arch measures), chi-square (prevalences), t-tests and chi-square for pacifier vs digit at equal duration; P < .05.

## 📏 How each measurement was defined (Figs 1–5)

| Measurement | Definition |
|---|---|
| Intercanine width | Primary canine cusp tip to cusp tip |
| Intermolar width | Mesiobuccal cusp tips of the second primary molars |
| Arch length | Right + left segments: anterior (central incisor contact to canine–first molar contact) + posterior (that contact to the distal of the second molar) |
| Arch depth | Perpendicular from the central incisor midpoint to a line joining the distal of the canines (canine depth) or of the second molars (molar depth) |
| Palatal depth | Deepest palatal point to a line joining the mesiolingual cusps of the second primary molars |
| Overjet | Horizontal distance, labial incisal edge of the lower central to the most labial upper central |
| Overbite | Vertical incisal-edge overlap (pencil line on the lower incisor); for open bite, the vertical gap |

## 📊 Breast-feeding (Tables I–II)

**No effect** on any of 15 arch measurements or 5 occlusal traits (every P ≥ .114) and no trend. Group sizes **48, 25, 33, 13**. Open bite 2.1 %, 0 %, 6.1 %, 15.4 % (P = .114) and crossbite 6.3 %, 8.0 %, 0 %, 15.4 % (P = .222) look higher after >12 months, but that group is only 13 children.

## 🍼 Pacifier (Tables III–IV)

| Outcome | <12 (n = 91) | 12–24 (44) | 24–36 (43) | 36–48 (18) | ≥48 (12) | P |
|---|---|---|---|---|---|---|
| Posterior crossbite % | 6.6 | 4.5 | 16.3 | 22.2 | **41.7** | .001 |
| Anterior open bite % | 5.5 | 4.5 | 0 | 0 | **25.0** | .008 |
| One or more traits % | 16.5 | 15.9 | 18.6 | 33.3 | **66.7** | .001 |
| Class II canine % | 33.0 | 20.5 | 34.9 | 44.4 | 50.0 | .210 NS |
| Mandibular canine width, mm | 22.6 | 22.6 | 22.5 | **23.6** | 23.4 | .041 |
| Overbite, mm | 1.9 | 2.1 | 2.0 | 1.7 | **0.4** | .007 |
| Palatal depth, mm | 14.8 | 14.3 | 14.3 | 13.9 | **13.3** | <.001 |
| Overjet, mm | 2.4 | 2.4 | 2.3 | 2.5 | 2.1 | .683 NS |

Crossbite comes from a **wider lower intercanine width** (only 36–48 months differed significantly; lower molar width P = .137) plus a nonsignificant narrowing of the upper (canine width 28.8 → 27.5 mm, P = .221). Overbite fell only at ≥48 months; palatal depth was shallower from 36 months. Overjet ≥4 mm did not change (P = .794).

## 👍 Digit (Tables V–VI)

| Outcome | <12 (n = 91) | 12–24 (36) | 24–36 (24) | 36–48 (13) | ≥48 (46) | P |
|---|---|---|---|---|---|---|
| Anterior open bite % | 5.5 | 0 | 0 | 15.4 | **37.0** | <.001 |
| Overjet ≥4 mm % | 6.6 | 0 | 8.7 | 15.4 | **39.1** | <.001 |
| One or more traits % | 16.5 | 13.9 | 8.7 | 23.1 | **71.7** | <.001 |
| Posterior crossbite % | 6.6 | 13.9 | 0 | 0 | 15.2 | .099 NS |

| Outcome | <12 (n = 91) | 12–24 (36) | 24–36 (24) | 36–48 (13) | ≥48 (46) | P |
|---|---|---|---|---|---|---|
| Class II canine % | 33.0 | 27.8 | 33.3 | 15.4 | 47.8 | .161 NS |
| Overjet, mm | 2.4 | 2.3 | 2.2 | **3.2** | **3.7** | <.001 |
| Overbite, mm | 1.9 | 2.0 | 2.1 | 1.6 | **0.6** | .006 |
| Maxillary canine width, mm | 28.8 | 29.2 | 29.0 | 29.0 | **27.8** | .009 |
| Maxillary molar width, mm | 41.5 | 42.0 | 42.1 | 41.2 | 40.5 | .037 |
| Maxillary canine depth, mm | 10.0 | 10.4 | 10.0 | 10.8 | **11.2** | <.001 |
| Maxillary molar depth, mm | 25.8 | 26.5 | 25.7 | **27.0** | **27.0** | .003 |

Digit habits make the upper arch **narrower and deeper**, not longer: arch length did not change with either habit (all P ≥ .193). Molar width at ≥48 months differed from the 12–36-month groups, not from <12 months. Palatal depth also differed (P = .030) but without a duration trend.

## 🔀 Pacifier vs digit

| | 🍼 Pacifier | 👍 Digit (thumb or finger) |
|---|---|---|
| Signature | **Posterior crossbite**: lower arch wider, upper slightly narrower | **Overjet**: upper arch narrower and deeper |
| Open bite | Yes, tends to be more symmetrical | Yes, not unusually **asymmetrical**, on the side of the hand used |
| Starts to matter | Beyond **24 months** | **36–48 months** |
| Head to head | More crossbite at 24–36 and ≥48 months (P = .034, .044); wider lower arch at 36–48 months (P = .013, t-test) | More overjet at ≥48 months (P = .001, t-test); more overjet ≥4 mm (P = .012) |

Pacifier means crossbite; digit means overjet; both mean open bite.

## 🦷 Clinical cases (Figs 6–8)

- **Fig 6, pacifier beyond 48 months:** slight open bite, **unilateral posterior crossbite**, overjet **1.5 mm** (also a fused left lateral and central incisor).
- **Fig 7, thumb beyond 48 months:** open bite, **Class II canines**, overjet **10 mm**.
- **Fig 8, thumb beyond 48 months: asymmetrical** open bite. Pacifier open bites are more symmetrical, probably because the pacifier's design limits where it sits.

## 💡 Discussion

- Most breast-fed children also suck nonnutritively; using the minimal-sucking group partly controls for this. Breast-feeding of **2 years or more** was not tested.
- Consistent with earlier work; limited mixed-dentition evidence (Bowden; Larsson) suggests increased overjet and narrow maxillary widths **persist**.
- Pacifier open bite was 0 % when the habit stopped between 24 and 48 months but 25 % when it reached 48 months (derived from Table IV): for pacifiers, open bite seems to need a longer habit while crossbite persists after stopping (interpretation; digit habits stopped at 36–48 months still showed 15.4 % open bite).
- AAO, ADA, AAP and AAPD advised concern only if habits reach the **early mixed dentition**; this may limit severity but may not cut prevalence as much as earlier cessation would.

## ⚠️ Limitations

- **Authors':** not a defined population; very few nonwhite children; habits parent-reported and unvalidated; small subgroups (breast-feeding >12 months n = 13; pacifier ≥48 months n = 12).
- Small numbers: 41.7 % crossbite = **5 of 12** children; 15.4 % digit open bite = **2 of 13** (derived).
- Habit intensity not recorded; blinding, reliability statistics and multiple-comparison correction not reported, post hoc test unnamed (Tables III and V show letter groupings); no confounder adjustment: **associations, not causes**.
- Reporting notes: the <12-month group (n = 91) is shared by Tables III and V, yet mandibular molar depth prints 22.2 mm vs 23.2 mm; the breast-feeding analysis uses 119 children, the shared reference group 91 (not explained).

## 🧠 Main conclusions

1. Breast-feeding length made no difference (in children with little nonnutritive sucking).
2. Both habits are linked to **open bite and reduced overbite**; **pacifier → posterior crossbite**; **digit → overjet with a narrow, deep upper arch**.
3. Some changes persisted well after the habit stopped: pacifiers stopped by 36 months still gave 16.3 % crossbite at age 5; digits stopped by 48 months gave 15.4 % open bite and 3.2 mm overjet.
4. Advice to worry only if a habit lasts into the early mixed dentition (AAO, ADA, AAP, AAPD) may be too late: habits beyond **2–3 years** clearly raise the risk.

| ✅ Number | Meaning |
|---|---|
| 372 | children analysed at 4.5–5 years (flow prints 547 − 98 − 56 = 393, not 372) |
| 41.7 % vs 6.6 % | posterior crossbite: pacifier ≥48 months vs <12 months |
| 16.3 % | crossbite at 5 years after a pacifier stopped by 36 months |
| 37.0 % / 39.1 % | open bite / overjet ≥4 mm with digit habits ≥48 months |
| 3.7 vs 2.4 mm | overjet: digit ≥48 months vs <12 months |
| 66.7 % / 71.7 % | at least one problem trait: pacifier / digit ≥48 months |
| 2–3 years | age after which habits clearly raise malocclusion risk |

# Myofunctional Therapy for Tongue-Thrusting: Background and Recommendations

by **Proffit and Mason** · J Am Dent Assoc 1975;90(2):403–411 · Review with the authors' pressure research, Florida and Kentucky

## 🎯 Purpose

To review how **form and function interact** in tongue thrust and give **guidelines for selecting patients for myofunctional (swallowing) therapy**. Built on the authors' intraoral pressure-transducer research at Kentucky (Fig 1), the literature and clinical experience.

- **The controversy:** advocates say tongue pressure pushes incisors into open bite and protrusion; many dentists wrongly believe therapy can **prevent** malocclusion, and refer children with **no speech problem** to speech clinicians, an ethical problem.

## 👅 What is tongue thrust? Three forms

No single name or definition (tongue-thrust swallow, visceral, infantile, reverse or deviant swallow, tongue-thrust syndrome). Cleall found these "abnormalities" in normal people. The authors' definition is one or a combination of:

| Form | What is seen (p 404) |
|---|---|
| **Swallow** | At the start of the swallow, tongue tip forward between the incisors, touching the lower lip |
| **Speech** | Tongue fronted against or between the incisors, jaw hinged open, in sounds not meant for that placement |
| **Rest** | Tongue forward, jaw slightly open, tip against or between the incisors |

All three occur in people **with and without** speech or dental problems.

## 🔁 Habit or normal development?

- **Straub** called it an "abnormal swallowing habit" caused by **bottle-feeding**: 478 orthodontic patients, only 2 breast-fed. Correlation, not cause: few variables, and Hanson and Cohen's bottle-feeding data do not support him.

| | 👶 Infant swallow | 👴 Adult swallow |
|---|---|---|
| Jaws | Apart; tongue fills the space between gum pads (later teeth) | Teeth brought together |
| Lips | Active in sucking | Relaxed, little activity |
| Tongue | Between the ridges, touching the lower lip, under the nipple | Tip on the palate behind the maxillary central incisors, stays there |

- **Bosma:** front-to-back maturation; lips mature before the tongue, and tongue-to-lower-lip contact persists through infancy.
- The adult pattern appears **between 2 and 12 years**; many normal children finish only near puberty, and it is not the pattern of all adults. The transition has been seen directly in longitudinal pressure-transducer studies and with cineradiography. Tongue thrust is usually a delayed normal stage, not a learned habit.

**Three other factors (p 405)**

1. **Thumb-sucking** delays the transition; the swallow will not change until sucking stops, and often changes spontaneously then. If not, therapy to teach the unlearned pattern may be indicated.
2. **Neuromotor problems** severe enough to block adult swallowing are rare; suspect them when general coordination is poor, with or without intellectual disability.
3. **Reversion** to a thrust after the adult swallow is learned is a true habit, like prolonged thumb-sucking; seen only in older individuals, and rare.

## 🫁 Why young children carry the tongue forward

- **Scammon curves (Fig 2):** the tongue grows like neural tissue (near full size by about **8 years**); the mandible follows the general body curve: slower, a plateau at **8–12 years**, then pubertal and postpubertal growth, some into the 20s. The relatively big tongue sits high and forward.
- **Airway:** the resting tongue position is set together with the airway. Lymphoid tissue grows fast, **peaks before puberty**, then regresses; tonsils and adenoids broadly follow this curve. Prominent tonsils are common at **8–10 years**: the tongue comes forward and the jaw hinges open to clear the airway; a sore throat reinforces the low, forward tongue.
- **Nasal obstruction** (adenoids, allergy, infection, turbinates, deviated septum) leads to mouth breathing. Ricketts' **respiratory obstruction syndrome** includes tongue thrust and malocclusion: tongue low and forward, mandible below its normal rest position. The key is to correct the breathing problem.
- **Orthognathic surgery:** setting the mandible back shrinks the oral cavity, yet the tongue does not block the airway; it moves **down and forward** and the hyoid takes a new position. Speech is unchanged; swallowing pressures usually fall, then stabilise at normal within a year. Speech adapts faster and more completely than swallowing; relapse from failed adaptation is uncommon.
- **Puberty:** oropharyngeal space grows quickly: the ramus grows (tongue shifts down), tonsils and adenoids involute, and the cervical vertebral bodies grow taller. The tongue rests further back and the arches come together; thrusters often reach an adult swallow **between 8 and 12 years** without therapy.
- **Sensory innervation:** denser nerve endings at the front encourage tongue fronting for tactile feedback; hence early bilabial and linguoalveolar sounds.

## 🔀 Three causes of tongue thrust in the mixed dentition

1. **Adaptation** to the mouth, pharynx and airway (and to tooth position).
2. **Neuromotor patterning** learned from feedback; with a neurological abnormality the prognosis for any therapy is poor.
3. A **normal but delayed transition**: common definitions, the authors' included, label normal stages as thrusting.

## 📊 Key evidence: prevalence

| Finding | Number |
|---|---|
| Anterior open bite, US children 6–11 years (USPHS, about 8,000) | **5.7 %** (black 16.3 %, white 3.9 %) |
| Open bite ≥2 mm | black 9.6 %, white 1.4 % |
| Tongue-thrust swallow (Fletcher; Hanson and Cohen similar) | about **50 %** at 6 years, **25 %** at 15 |
| Thumb-sucking (USPHS) | about 10 % |

| Finding | Number |
|---|---|
| Simple open bites that self-correct (1,400 Navajo children, cross-sectional; Worms) | **80 %** |
| Excess maxillary incisor protrusion | about 17 %: more than thumb-sucking, so not mainly thumb-sucking; genetics a significant cause |
| Myofunctional therapy, controlled studies (Subtelny) | does **not** significantly change the swallow pattern |

- **Tongue thrust and thumb-sucking are far more common than the open bite they are said to cause.** Association, not cause and effect.
- Fig 3: an open bite closing over **4 years** without therapy. Worms: in 7–9-year-old Navajo boys a therapy would have to succeed in **more than 80 %** to beat maturation.
- Protrusion is strongly linked to thumb-sucking that continues after the permanent incisors erupt; any link with tongue thrust **cannot be determined** from the data.

## ⚖️ Pressure studies: the equilibrium question

Tongue and lip pressures during swallowing are **not balanced**, even over long periods, and do not seem to shape the arches strongly.

| Study (p 408–409) | Finding | What it shows |
|---|---|---|
| Australian aborigines vs Kentucky whites (Fig 4) | Swallowing, maxillary lingual (n = 16 each): about 90–120 vs 165–225 g/cm²; resting, mandibular lingual (n = 13 vs 8): about 10–18 vs 13–25 g/cm² (read from the figure) | **Larger** aboriginal arches, **lower** tongue pressure |
| Protruded incisors | **Less** tongue pressure; back to normal after retraction | Pressure follows tooth position |
| Resting lip pressure | Stronger relation with arch form | Rest matters more than function |
| Wallen, open bite vs normal | Vertical tongue forces **lower** in open bite | The opposite of the swallowing theory |
| Eruption force | Only about **5 g** | Light resting forces could matter |

**Resting posture matters more than swallowing:** a forward resting tongue may impede incisor eruption while posterior teeth keep erupting, giving an open bite, though it was **never shown directly**. Any tongue effect on malocclusion works through resting posture and eruption, not the swallow.

## 🦷 Recommendations by clinical scenario

| Situation | What to do |
|---|---|
| Tongue thrust only | No therapy; it has no preventive role. No evidence that speech problems or open bite will develop; the child will almost surely mature on their own |
| With a speech problem | Do not assume one causes the other. Only a speech therapist, using articulation therapy on the usual speech-clinic grounds; swallowing work alongside it before puberty is usually unnecessary and contraindicated. Articulation therapy also moves the tongue tip back at rest |
| With malocclusion | Treat the malocclusion first: without obvious predisposing factors the thrust disappears in about **80 %** (a distillation of orthodontic experience) |
| Malocclusion and speech problem, puberty or later | Speech therapy alongside open-bite orthodontics, never delayed until it starts; resting-posture exercises and adaptive phonetic placements; earlier orthodontics may ease tongue placement |
| Still present at puberty | Therapy for the **resting** tongue position, **combined with orthodontics** |
| Small airway (reduced faucial isthmus) | No swallowing exercises; adenotonsillectomy decided by a physician on physical complaints; then allow time for an adult swallow |

**Why postpone tongue therapy until orthodontics (p 409–410):** 1. correction usually removes the thrust; 2. the child gets the most time to mature, and school-age swallowing is highly variable; 3. therapy works best during tooth movement, as function adapts to form. Orthodontics may cause a **temporary** loss of speech skills; most children adapt quickly.

## ✅ The five recommendations (p 411)

1. **No therapy** without a malocclusion or speech problem: there is nothing to prevent.
2. **No swallowing therapy before puberty**, even with malocclusion (maturational age matters more than chronological age); if the thrust persists into puberty, therapy to change tongue position is probably indicated.
3. Tongue therapy works best **combined with orthodontics**.
4. Lisp with thrust or malocclusion before puberty: **speech therapy now**; classic myofunctional therapy adds little and is contraindicated with a small faucial isthmus.
5. Older children with speech problems under open-bite treatment: myofunctional techniques may help modify the anterior **resting** posture; articulation therapy helps move the tongue tip back.

## ⚠️ Limitations

Narrative review leaning on the authors' own laboratory. The three **80 %** figures come from different sources: Navajo cross-sectional data, a distillation of orthodontic experience, and the summary's **"up to 80 %"** improving by 12; none compares treated and untreated cohorts.

| ✅ Number | Meaning |
|---|---|
| 2–12 years | when the adult swallow develops |
| 5.7 % | anterior open bite in US children aged 6–11 (USPHS) |
| 50 % → 25 % | tongue-thrust swallow at 6 vs 15 years (Fletcher) |
| 10 % vs 17 % | thumb-sucking (USPHS) vs excessive maxillary incisor protrusion (p 408) |
| 80 % | simple open bites self-correcting in Navajo children |
| 5 g | force of tooth eruption |
| up to 80 % | children with tongue thrust and open bite at 8 who improve by 12 without therapy |
| 478 / 2 | Straub's orthodontic patients / of whom breast-fed: correlation only |